+5V

1x 100nF per VDD Pins

+3V3

C1 100nF
C3 100nF
C4 100nF
C6 100nF
C7 100nF
C9 100nF
C20 100nF
C11 2uF2

GND

U7C 74HC74

C13 100nF

VCC 14
GND 7

U7B 74HC74

D 12
C 11
S 10
R 13
Q 9
Q̄ 8

COM1_INPUT

J2 HDMI_A_1.4

D2+ 1 HDMI_1_D2+
D2− 3 HDMI_1_D2−
D1+ 4 HDMI_1_D1+
D1− 6 HDMI_1_D1−
D0+ 7 HDMI_1_D0+
D0− 9 HDMI_1_D0−
CK+ 10 HDMI_1_CK+
CK− 12 HDMI_1_CK−
CEC 13 HDMI_1_CEC
SCL 15 HDMI_1_SCK
SDA 16 HDMI_1_SDA
UTILITY/HEAC+ 14 HDMI_1_UTIL
HPD/HEAC− 19 HDMI_1_HPD

SH 18
+5V 18
GND 17
CKS 11
DOS 8
D1S 5
D2S 2

GND

HDMI_MUX

HDMI_1_D2+ HDMI_1_OUT_CK+ HDMI_1_OUT_Ck+
HDMI_1_D2− HDMI_1_OUT_CK− HDMI_1_OUT_CK−
HDMI_1_D1+ HDMI_1_OUT_D0+ HDMI_1_OUT_D0+
HDMI_1_D1− HDMI_1_OUT_D0− HDMI_1_OUT_D0−
HDMI_1_D0+ HDMI_1_OUT_D1+ HDMI_1_OUT_D1+
HDMI_1_D0− HDMI_1_OUT_D1− HDMI_1_OUT_D1−
HDMI_1_CK+ HDMI_1_OUT_D2+ HDMI_1_OUT_D2+
HDMI_1_CK− HDMI_1_OUT_D2− HDMI_1_OUT_D2−
HDMI_1_CEC
HDMI_1_UTIL
HDMI_1_HPD
HDMI_COM_SELECT

File: HDMI_MUX.kicad_sch

+3V3

RST1 SW_Push
C2 100nF
R8 10K
NRST
SW1 SW_SPDT
R9 10K
TP4 TestPoint
BOOT0

GND

MCU_OSC_IN
MCU_OSC_OUT
Y1 Crystal
C5 10pF
16MHz
C8 10pF

GND

GRN LED
RED LED
BLU LED

R11 33
HDMI_COM_SELECT

C10 2uF2
C12 2uF2

U1 STM32F411VETx

VBAT 6
VDD 11
VDD 19
VDD 28
VDD 50
VDD 75
VDD 100
VDD 22
VDDA
VSS 10
VSSA 20

NRST 14
BOOT0 94
VREF+ 21
RCC_OSC_IN 12
RCC_OSC_OUT 13

PE0 97
PE1 98
PE2 1
PE3 2
PE4 3
PE5 4
PE6 5
PE7 38
PE8 39
PE9 40
PE10 41
PE11 42
PE12 43
PE13 44
PE14 45
PE15 46

PD0 81
PD1 82
PD2 83
PD3 84
PD4 85
PD5 86
PD6 87
PD7 88
PD8 55
PD9 56
PD10 57
PD11 58
PD12 59
PD13 60
PD14 61
PD15 62

VCAP1 48
VCAP2 73

PA0 23
PA1 24
PA2 25
PA3 26
PA4 29
PA5 30
PA6 31
PA7 32
PA8 67
PA9 68
PA10 69
USB_OTG_FS_DM 70
USB_OTG_FS_DP 71
SYS_JTMS−SWDIO 72
SYS_JTCK−SWCLK 76
PA15 77

PB0 35
PB1 36
PB2 37
PB3 89
PB4 90
PB5 91
I2C1_SCL 92
I2C1_SDA 93
PB8 95
PB9 96
PB10 47
PB12 51
PB13 52
PB14 53
PB15 54

PC0 15 HDMI_COM_SEL
PC1 16 HDMI_MUX_EN
PC2 17 HDMI_TCA_EN
PC3 18
PC4 33
PC5 34
TIM3_CH1 63
PC7 64
TIM3_CH3 65
TIM3_CH4 66
PC10 78
PC11 79
PC12 80
PC13 7
PC14 8
PC15 9

+3V3

R14 5K
R15 5K

5V

C21 100nF
C19 100nF

GND

R17 2K
R18 2K

VCCA 1
VCCB 8
SCLA 2
SDAA 3
SCLB 7
SDAB 6
GND 4
EN 5 HDMI_TCA_EN

U6 TCA9517ADGK

GND

uC_I2C1_SCL
uC_I2C1_SDA

+5V

MONITOR_HDMI_OUT

J3 HDMI_A_1.4

HDMI_1_OUT_D2+ 1 D2+
HDMI_1_OUT_D2− 3 D2−
HDMI_1_OUT_D1+ 4 D1+
HDMI_1_OUT_D1− 6 D1−
HDMI_1_OUT_D0+ 7 D0+
HDMI_1_OUT_D0− 9 D0−
HDMI_1_OUT_Ck+ 10 CK+
HDMI_1_OUT_CK− 12 CK−
HDMI_1_OUT_CEC 13 CEC
HDMI_1_OUT_SCK 15 SCL
HDMI_1_OUT_SDA 16 SDA
HDMI_1_OUT_UTIL 14 UTILITY/HEAC+
HDMI_1_OUT_HPD 19 HPD/HEAC−

D2S 2
D1S 5
DOS 8
CKS 11
GND 17
SH 14

EDID_EMULATION

COM1
HDMI1_SCL
HDMI1_SDA

HDMI_1_SCK
HDMI_1_SDA

EDID PROGRAM
uC_I2C1_SCL
uC_I2C1_SDA

HDMI1_OUT_SCK HDMI_1_OUT_SCK
HDMI1_OUT_SDA HDMI_1_OUT_SDA
COM1 OUT

HDMI_COM_SEL
HDMI_MUX_EN

uC_COM_SELECT
uC_COM_HDMI_MUX_EN

File: EDID_EMULATION.kicad_sch

TCA9517ADGK Notes:
B SIDE This means that the VOL of any
device on the B−side must be below 0.45V

R16 10K

## DEBUG LEDS

GRN LED R3 500 D1 GRN LED
RED LED R4 500 D2 RED LED
BLU LED R5 500 D3 BLU LED

GND

## COMPUTER SELECT BUTTON

COM_SELECT = LOW → Computer 1 (Port A)
COM_SELECT = HIGH → Computer 2 (Port B)

+3V3

R2 10K
R1 10K

SW2 COM1_SELECT

U8A 74HC7014

U7A 74HC74

COM_SELECT

R6 47k
C14 470nF
tau=~22(ms)

POR?

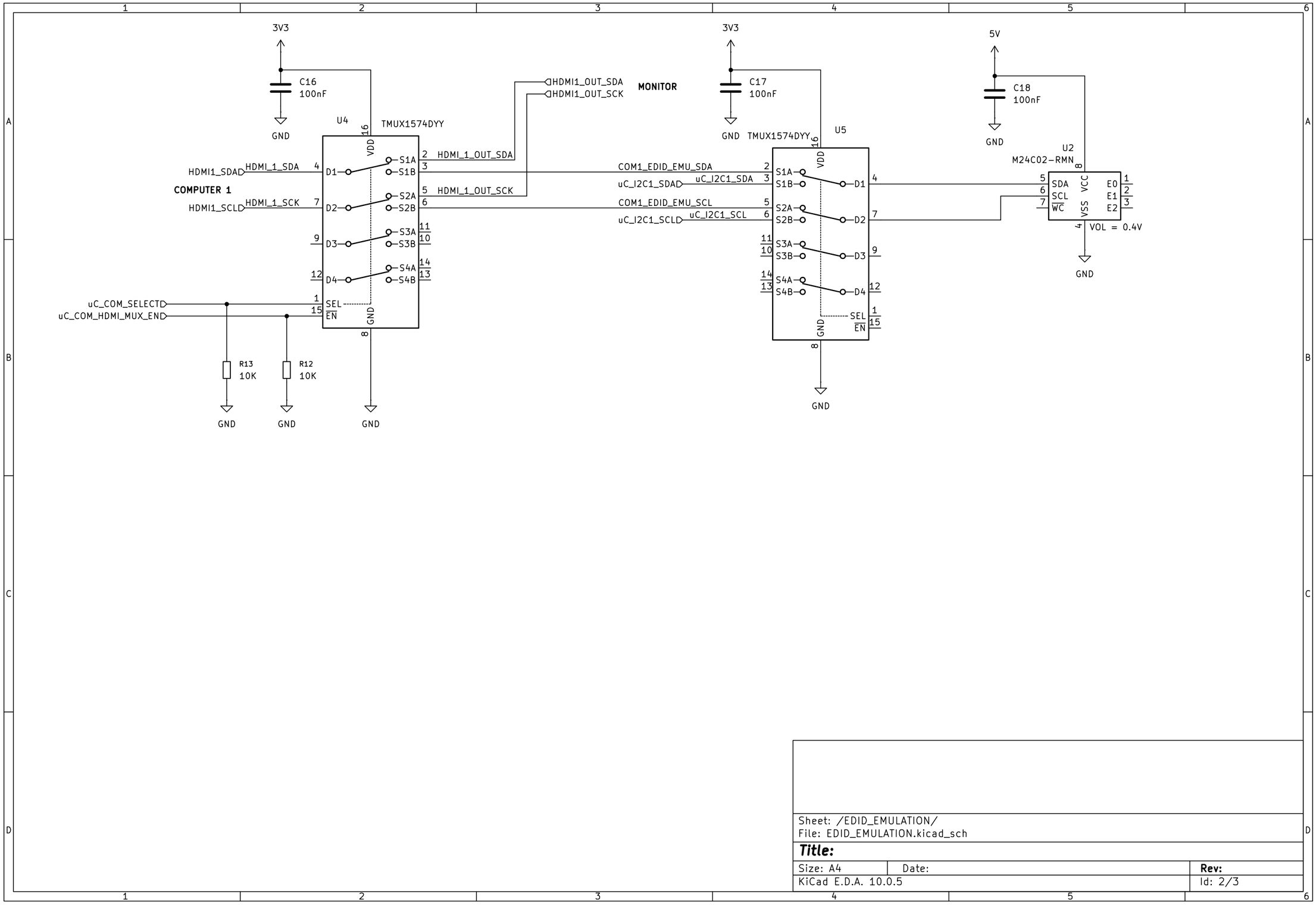
GND

Q = HIGH = COM1
Q = LOW = COM2

Sheet: /
File: openKVM−X01.kicad_sch

**Title: Open KVM**

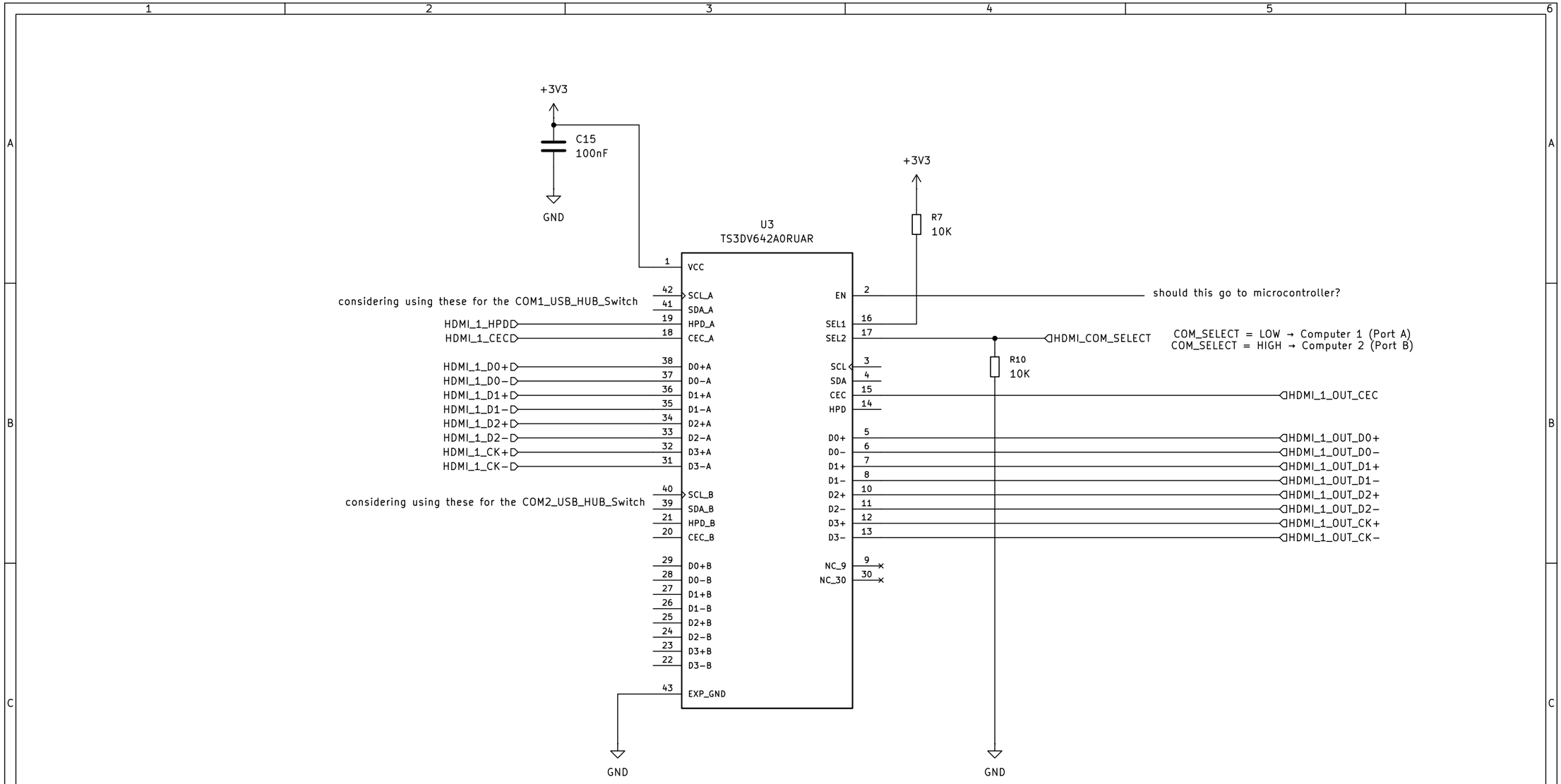
Size: B | Date: | **Rev: X01**
KiCad E.D.A. 10.0.5 | Id: 1/3




Sheet: /EDID_EMULATION/
File: EDID_EMULATION.kicad_sch

***Title:***

| Size: A4 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 2/3 |




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|---|---|---|
| Sheet: /HDMI_MUX/<br>File: HDMI_MUX.kicad_sch | | |
| ***Title:*** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 3/3 |